Probability Theory - 01040222

Homework 1

Yonatan Avidor - 214269565
Rotem Eshet - 212674683

Question 1 - Words over a Three-Letter Alphabet

Let $n \geq 3$. A word of lenth n is chosen uniformly at random from the alphabet $\{A, B, C\}$. In other words, all 3^n possible elements of $\{A, B, C\}^n$ have equal probability $\frac{1}{3^n}$

Section A

What is the probability that the letter A does not appear in the word?

Section B

What is the probability that exactly two distinct letter appear in the word?

Section C

What is the probability that all three letters appear in the word? Find the limit of this probability as $n \to \infty$

Answer 1

Section A

First, we'll define our sample space Ω to be $\{A, B, C\}^n$.
We'll define U as

$$U = \{x \in \Omega : A \text{ does not appear in } x\}$$

Since the distribution is uniform, $\mathbb{P}(U) = \frac{|U|}{|\Omega|}$. We know that $|\Omega| = 3^n$, now we only need to find $|U|$.
For A not to appear in x, each letter must be either B or C, so there are 2^n possible words in U. Therefore

$$\mathbb{P}(U) = \frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n$$

Section B

We'll define Q as

$$Q = \{x \in \Omega : \text{Exactly two distinct letters appear in } x\}$$

Notice that there are only three words that have only one letter, they are $\{A\}^n, \{B\}^n, \{C\}^n$.
From section A, we know that there are 2^n words that do not contain A, we need to subtract 2 from this amount (to consider the words that have only one letter) to get the words that have both B and C but not A.
Then, from symmetry we can multiply this number by 3 and get $|Q|$.
As mentioned in Section A, since the probability is uniform we have

$$\mathbb{P}(Q) = \frac{|Q|}{|\Omega|} = \frac{3(2^n - 2)}{3^n} = \boxed{\frac{2^n - 2}{3^{n-1}}}$$

Section C

Let $n \in \mathbb{N}$
We'll define V as

$$V = \{x \in \Omega : \text{At least one letter does not appear in } x\}$$

And we want to find $\mathbb{P}(V^c)$

V is the set of words that have exactly two distinct letters OR one distinct letter, Q is the former while $\{\{A\}^n, \{B\}^n, \{C\}^n\}$ is the latter. Therefore $|V| = |Q| + 3$,

$$\mathbb{P}(V^c) = 1 - \mathbb{P}(V) \underset{\text{uniform prob.}}{=} 1 - \frac{|Q| - 3}{|\Omega|} = 1 - \frac{2^n - 1}{3^n} \xrightarrow{n\to\infty} 1 - 0 = \boxed{1}$$

Question 2 - A Countably Infinite Probability Space

Consider the sample space $\Omega = \{1, 2, 3, ...\}$, with probability function

$$p(k) = \frac{c}{3^k}$$

Where $c > 0$ is a constant

Section A

Find c such that (Ω, p) is a discrete probability space

Section B

Calculate the probability that the chosen integer is even.

Section C

For every integer $m \geq 1$, calculate the probability that the chosen integer is at least m. Find the limit of this probability as $m \to \infty$.

Answer 2

Section A

For (Ω, p) to be a discrete probability space, we need that $\mathbb{P}(\Omega) = 1$.
So we need to find $c > 0$ such that

$$\sum_{k=1}^{\infty} \frac{c}{3^k} = 1$$

$$\sum_{k=1}^{\infty} \frac{c}{3^k} = c\sum_{k=1}^{\infty} \frac{1}{3^k} \underset{\text{Infinite Geometric Series}}{=} c\frac{\frac{1}{3}}{1-\frac{1}{3}} = c\frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}c$$

$$\implies \frac{1}{2}c = 1 \implies \boxed{c = 2}$$

Section B

Since we are working with a discrete probability space, the probability of getting an even number equals the sum of probabilities for all even numbers. we therefore add the probabilities for all even n's:

$$\mathbb{P}(\text{getting an even number}) = \sum_{n=1}^{\infty} \frac{2}{3^{2n}} = \sum_{n=1}^{\infty} \frac{2}{9^n} \underset{\text{Infinte Geometric Series}}{=} \frac{2}{8} = \boxed{\frac{1}{4}}$$

Section C

Since we are in a discrete probability space, the probability of getting a number at least m is the sum of probability for each $k \geq m$

$$\sum_{k=m}^{\infty} p(k) = \sum_{k=m}^{\infty} \frac{2}{3^k} = \frac{\frac{2}{3^m}}{\frac{2}{3}} = \frac{1}{3^{m-1}} \overset{m\to\infty}{\longrightarrow} \boxed{0}$$

Question 3 - Two Random Subsets

Let $n \geq 1$. An ordered pair of subsets (A, B) of $\{1, .., n\}$ is chosen uniformly at random. In other words, our sample space is

$$\Omega = \{(A, B) : A, B \subseteq \{1, ..., n\})$$

And all 4^n outcomes are equally likely.

Section A

What is the probability that $A \cap B = \varnothing$?

Section B

What is the probability that $A \subseteq B$?

Section C

What is the probability that $A \subseteq B$ or $B \subseteq A$?

Section D

For $0 \leq k \leq n$, what is the probability that $|A \cap B| = k$?

Answer 3

First, notice that for for every element in $x \in [n]$, either $x \in A$ or (exclusive!) $x \notin A$, and also $x \in B$ or (still exclusive!) $x \notin B$, and since the probability is uniform, for each of those situations the probability is $\frac{1}{2}$. Furthermore, the

$$\mathbb{P}(x \in A \wedge x \in B) = \mathbb{P}(x \notin A \wedge x \notin B)$$

$$= \mathbb{P}(x \in A \wedge x \notin B) = \mathbb{P}(x \notin A \wedge x \in B) = \frac{1}{4}$$

Section A

For the intersection to be empty, we need that

$$\forall x \in [n] \quad (x \notin A \wedge x \notin B) \vee (x \in A \wedge x \notin B) \vee (x \notin A \wedge x \in B)$$

And from what we've established

$$\mathbb{P}((x \notin A \wedge x \notin B) \vee (x \in A \wedge x \notin B) \vee (x \notin A \wedge x \in B)) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$$

Since this is true for all $x \in [n]$, we get $\boxed{\left(\frac{3}{4}\right)^n}$.

Section B

For A to be a subset of B, we need that

$$\forall x \in [n] \quad (x \notin A \wedge x \notin B) \vee (x \in A \wedge x \in B) \vee (x \notin A \wedge x \in B)$$

So just like in section, we have

$$\mathbb{P}(x \notin A \wedge x \notin B) \vee (x \in A \wedge x \in B) \vee (x \notin A \wedge x \in B) = \boxed{\left(\frac{3}{4}\right)^n}$$

Section C

Since the probability is uniform in a discrete sample space, we need to calculate the number of possibilies.
We will use the Principle of Inclusion and Exclusion.

$$|A \subseteq B \cup B \subseteq A| = |A \subseteq B| + |B \subseteq A| - |\underbrace{A \subseteq B \cap B \subseteq A}_{(A=B)}|$$

From section B and symmetry we know that $\mathbb{P}(A \subseteq B) = \mathbb{P}(B \subseteq A) = \left(\frac{3}{4}\right)^n$
To get $A = B$ we need for all $x \in [n] \quad (x \in A \wedge x \in B) \vee (x \notin A \wedge x \notin B)$ So $\mathbb{P}(A = B) = \left(\frac{1}{2}\right)^n$
Therefore $\mathbb{P}(A \subseteq B \cup B \subseteq A) = \boxed{2\left(\frac{3}{4}\right)^n - \left(\frac{1}{2}\right)^n}$

Section D

We need k elements to be in A and in B, there are $\binom{n}{k}$ options for said k elements. Then, for each of the other $n-k$ elements, they need to either be in A but not in B, be in B but not in A or to not be in either. Therefore there are 3^{n-k} options for them overall.
Since the probability is uniform, The probability is

$$\mathbb{P}(|A \cap B\ | = k) = \frac{|\{(A,B) \in \Omega : |A \cap B| = k\}|}{|\Omega|} = \boxed{\frac{\binom{n}{k} \cdot 3^{n-k}}{4^n}}$$

Corny Joke

Pikachu! $\binom{\text{I}}{\text{You!}}$
(If I write combinatorics answers, I also write combinatorics jokes)